

Name:

Points:

Linux Server & Bash Programming — Final Exam Retake

2025-01-22

Instructions

1. No computers, no phones, no notes
2. No talking with other people in the room
3. In case point 1 or 2 are broken, you get automatically **0 points** on the test.
4. Write your name on top of the page (do not fill in the *Points* box!)
5. For each question, you'll need to answer the question by crossing with an **X** the right answer.
6. Once you're done:
 - Double-check your answers
 - Hand over your test to me
 - Leave the room

NB: While inside the classroom, rule 1 and 2 are still valid, even if you've already handed over your answers to me. If you want to discuss your answers with a classmate, or to check an answer on your phone, do that **outside the classroom**.

NB: Make sure to use the restroom, fill our water bottle, grab a coffee, etc... **before** we begin the exam, as you're not allowed to leave the classroom without handing over your answers.

Point system

Each question gives **1 point**. There are **20 questions**, for a total of maximum **20 points**.
Each correct answer gives **1 point**.

Grading

Points	Grade
0 - 9	IG
10 - 17	G
18 - 20	VG

READ EACH QUESTION CAREFULLY!

1 In Linux everything is...

- 1. Difficult
- 2. More fun
- ✓ 3. A file
- 4. A terminal
- 5. A shell

2 How many addresses can I have in a /28 network? (NB: addresses, not hosts)

- 1. 2^4
- 2. $2^4 - 2$
- ✗ 3. 2^5
- 4. $2^5 - 2$
- 5. 2^6
- 6. $2^6 - 2$

3 How can I access another machine's localhost?

- 1. By adding an entry to `/etc/hosts`
- 2. By creating or modifying a file in `/etc/netplan/`
- 3. It's not possible
- ✗ 4. By opening a listening socket in the target machine and accepting connections

4 DNS is...

- 1. A service that translates hostnames into IP addresses
- ✗ 2. A service that translates IP addresses into hostnames

5 A docker container is...

- 1. Like a variable, but global
- 2. Like a script, but better
- ✓ 3. A kind of virtual machine
- 4. An isolated process

6 What type of packages are used by Fedora?

☒ 1. apt

☐ 2. deb

3. yum

☒ 4. rpm

5. dnf

7 When I create a new user, the following file will be changed

☐ 1. /etc/users

☒ 2. /etc/passwd

3. /etc/shadow

4. /etc/light

8 A user on a Linux system can only be part of up to 2 groups: a group with the same name as the user and one additional group

☒ 1. true

☐ 2. false

9 When setting special permissions using the octal representation, the following value mappings are used:

☒ 1. SUID=4, SGID=2, sticky=1

☐ 2. SUID=4, sticky=2, SGID=1

3. SGID=4, SUID=2, sticky=1

☒ 4. SGID=4, sticky=2, SUID=1

10 How can I make sure the service X is started on boot?

☐ 1. systemctl start x.service

☒ 2. systemctl start --at-boot x.service

☒ 3. systemctl enable x.service

4. systemctl start --on-boot x.service

11 The network model in use today is called:

- 1. OSI Model
- ✓ ② TCP/IP Model
- 3. UDP/IP Model

12 Is it possible for two different sockets to bind to the same IP address?

- ✓ ① yes
- 2. no

13 In the "3-2-1" rule for backups, the "2" part refers to...

-
- 1. 2 different mediums
 - ✗ ② 2 different locations
 - 3. always take at least 2 copies
 - 4. always store at least 2 copies on a different location

14 When building docker images, it's possible to use the cache from a previous build

- ✓ ① true
- 2. false

15 A docker image is:

- ✓ 1. another name for a container
- 2. the text file used to build a container, usually called **Dockerfile**
- 3. connected to a tag, which can never change
- ④ the blueprint used to create containers

16 When starting a docker container from the cli (using **docker run**), I can create a bound volume using a relative path

- ✗
- ① true
 - 2. false

17 When exposing ports from a docker container to the host, I have to use the same port on both the host and the container

✓ 1. true

② false

18 To achieve IPC (inter-process communication) on Linux, I can use signals, sockets and...

→ 1. pipes

✗ ② redirects

3. processes

4. disk partitions

19 The GNU project is...

1. An ancestor of Linux

2. The original implementation of UNIX

✗ ③ A spin-off of Linux

→ 4. A new, FOSS implementation of the UNIX system

5. None of the above

20 The following git commands need a remote to be set to properly work:

1. git open and git close

→ 2. git pull and git push

✗ ③ git pull, git push and git remote

4. git add, git commit, git push, git pull and git remote